

Natural Science

I. Nature and Scope of Science

What is Science?

- Comes from the Latin term *scientia* which means "to know"
- A "body of knowledge" that grows with time as human learn more and more about principles or laws that govern the natural phenomenon

What are the different divisions of Science?

Science is divided into three major divisions which are:

- **Natural Sciences** - the study of the natural world
- **Social Sciences** - the systematic study of human behavior and society
- **Applied Sciences** - scientific knowledge is put to practical use

Natural Sciences:

- **Physical Sciences** - focus on the nature and behavior of matter and energy.
- **Earth Sciences** - examine the structure and composition of our planet, and the physical processes that have helped to shape it.
- **Life Sciences (Biology)** - study of the development, distribution, evolution, function, origin, and structure, of living things.

How do scientists search for an answer?

Scientists keep on asking questions about nature to explain phenomena, to search for truth, to formulate new concepts and theories and to solve some problems. In doing so a certain method can be used – the scientific method.

What is a Scientific Method?

- It is a systematic way to answer scientific questions and present the results to the scientific community.
- It is the logical way in which a scientist goes about trying to solve a problem.
- It is systematic way to study a research problem or issue

Steps of the Scientific Method

1. Identifying and clearly defining the problem

- The problem is the scientific question to be solved. It is best expressed as an "open-ended" question, which is a question that is answered with a statement, not just a yes or a no.

- Examples:
 - How does light affect the reproduction of bread mold on white bread?
 - What is the effect of fertilizer in the growth of the plants?
 - How does amount of water effect plant growth?
- 2. Gathering of preliminary data or pertinent preliminary information about the problem
- 3. Formulating a hypothesis on how to solve the problem
 - Two things must be kept in mind while stating a hypothesis:
 - It must be possible to measure the terms in the hypothesis
 - It must answer the original question
- 4. Designing an experiment to test the hypothesis and subsequently performing it
 - It is used to test the hypothesis. It has the following parts or components:
 - **Variables:** Element or elements being tested in the experiment
 - **Control:** The control is a particular sample that is treated the same as all the rest of the samples except that it is not exposed to manipulated variables.
 - A controlled experiment should be designed with the following variables:
 - **Manipulated (or Independent) Variable:** This is the variable that is changed in an experiment. For example, one wanted to know how fertilizer affects plant growth, the amount of fertilizer applied is the only variable that she would change. A good experiment will test more than one amount of fertilizer and compare the growth among all groups.
 - **Responding (or Dependent) Variable:** This is the variable that changes as a result of the changes in the manipulated variable. In the above example, the responding variable would be the size of the plants.
 - **Controlled Variables (or Controls):** These are all the things that will be kept the same in the experiment. Controls in the above example would include: the origin of the seeds, the amounts of light and water each plant receives, the type of soil used the size of the pot, and the temperature.
- 5. Analyzing and interpreting the experimental data
- 6. Drawing of generalization or conclusion
- 7. Verifying conclusion by applying it to similar situations.

Scientific Attitudes

It is a disposition or demonstration of feelings and/or thoughts the scientist shows while doing a research or investigation. The following are the traits most commonly ascribed to scientists:

1. Objectivity
 - considers all pros and cons
 - considers all evidences available
 - considers and evaluates statements by others
2. Critical-mindedness
 - looks for inconsistencies
 - consults a number of authorities
 - challenges the validity of statements
3. Suspended judgment
 - recognizes the restrictions in generalizations and theories
 - generalizes only to the degree justified by available evidence



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4. Open-mindedness

- considers several possible options when investigating a problem
- considers and evaluates ideas presented by others

5. Curiosity

- looks for inconsistencies
- challenges the validity of unsupported statements
- asks many questions starting with who, where, when and how

6. Patience

- waits for results of the experiments that may take years to accomplish

7. Perseverance

- continuously works on experiments that may not be successful right away

8. Accepting authority

- uses the findings of earlier scientists as basis of their experiments

9. Truthfulness

- states things with some degree of uncertainty is indeed the results of an investigation are not conclusive
- does not manufacture or manipulates data in order to prove the preferred result

10. Humility

- accepts the views and opinions of others
- admits error if there is
- realizes the overwhelming magnificence of creation

II. Life Sciences (Biology)

What is Biology?

- The term biology is derived from *bios* meaning life and *logos* meaning study
- Biology literally means "the study of life".

Branches of Biology

Biology is such a broad field, covering the minute workings of chemical machines inside our cells, to broad scale concepts of ecosystems and global climate change. The following are some of the branches of Biology:

- **Anatomy** - the study of the animal form, with an emphasis on human bodies
- **Biochemistry** - the study of the chemical reactions required for life to exist and function, usually a focus on the cellular level
- **Biotechnology** - a new and sometimes controversial branch of biology that studies the manipulation of living matter, including genetic modification
- **Botany** - the study of plants
- **Cell Biology/Cytology** - the study of the cell as a complete unit, and the molecular and chemical interactions that occur within a living cell.



- **Ecology** - the study of the ecosystem as a complete unit, with an emphasis on how species and groups of species interact with other living beings and non-living elements.
- **Entomology** - study of insects
- **Genetics** - the study of genes and heredity.
- **Histology** - The study of cells and tissue, a microscopic branch of anatomy.
- **Ichthyology** - the study of fish
- **Microbiology** - the study of microscopic organisms (microorganisms) and their interactions with other living things
- **Mycology** - the study of fungi
- **Ornithology** - the study of birds
- **Paleontology** - the study of fossils and sometimes geographic evidence of prehistoric life
- **Pathobiology or pathology** - the study of diseases, and the causes, processes, nature, and development of disease
- **Parasitology** - the study of parasites and parasitism
- **Physiology** - the study of the functioning of living organisms and the organs and parts of living organisms
- **Taxonomy** - the study of the naming and classification of living things
- **Virology** - the study of viruses and some other virus-like agents, usually considered part of microbiology or pathology
- **Zoology** - the study of animals and animal life, including classification, physiology, development, and behavior

Characteristics of Living Things

Complex organization

- Living things have a level of complexity and organization not found in lifeless objects.
- A living thing is composed of one or more **cells**. These units, generally too small to be seen with the unaided eye, are organized into tissues.
- A *tissue* is a series of cells that accomplish a shared function.
- Tissues, in turn, form *organs*, such as the stomach and kidney.
- A number of organs working together compose an *organ system*.
- An organism is a complex series of various organ systems.

Metabolism

- Living things exhibit a rapid turnover of chemical materials, which is referred to as **metabolism**.
- Metabolism involves exchanges of chemical matter with the external environment and extensive transformations of organic matter within the cells of a living organism.
- Metabolism generally involves the release or use of chemical energy.

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Responsiveness

- All living things are able to respond to stimuli in the external environment. For example, living things respond to changes in light, heat, sound, and chemical and mechanical contact.
- To detect stimuli, organisms have means for receiving information, such as eyes, ears, and taste buds.

Growth

- It refers to an increase in size.
- Growth requires an organism to take in material from the environment and organize the material into its own structures. To accomplish growth, an organism expends some of the energy it acquires during metabolism.

Reproduction

- A living thing has the ability to produce copies of itself by the process known as *reproduction*.
- Among plants and simple animals, reproduction is often an extension of the growth process. For example, bacteria grow and quickly reach maturity, after which they split into two organisms by the process of *asexual reproduction*.
- Asexual reproduction involves only one parent, and the resulting cells are generally identical to the parent cell.
- More complex organisms engage in a type of reproduction called *sexual reproduction*, in which two parents contribute to the formation of a new individual.
- During this process, a new combination of traits can be produced. The process is generally more complex than asexual reproduction, requiring that parents find one another, then (usually) care for their offspring before it can live independently.

Evolution

- Populations of living organisms have the ability to adapt to their environment through the process of evolution.
- During evolution, changes occur in populations, and the organisms in the population become better able to metabolize, respond, and reproduce.
- They develop abilities to cope with their environment that their ancestors did not have.
- Evolution also results in a greater variety of organisms than existed in previous eras. This proliferation of populations of organisms is unique to living things.



Homeostasis.

- Homeostasis is the maintenance of a constant (yet also dynamic) internal environment in terms of temperature, pH, water concentrations, etc.

Adaptation.

- Refers to modifications that promote the likelihood of survival
- There are two types of adaptations – behavioral and structural adaptations
- Structural adaptation refers to the body parts of organisms which enable them to adjust to the conditions in the environment. Examples are: thorn-like leaves of cacti plants, presence of prop roots among mangrove plants, thick fur of the polar bears, more stomates in the undersurface of the leaf
- Behavioral adaptation refers to what the organisms do to adjust to the conditions in the environment. Examples are: trees shedding leaves during summer, migration of birds, hibernation

Biology of Cells

- All living things are composed of cells.
- Some organisms consist of a single cell, while others have multiple cells
- One of the first scientists to observe cells was the Englishman Robert Hooke. Hooke examined a thin slice of cork through the newly invented microscope. The microscopic compartments in the cork impressed him and reminded him of rooms in a monastery, known as cells. He therefore referred to the units as **cells**.
- Anton Van Leeuwenhoek, a Dutch merchant, made further observations of plant, animal, and microorganism cells.
- In 1838, the German botanist Matthias Schleiden proposed that all plants are composed of cells.
- Theodore Schwann, concluded that all animals are also composed of cells.
- In 1858, the biologist Rudolf Virchow proposed that all living things are made of cells and that all cells arise from preexisting cells. These premises have come down to us as the *cell theory*.

Parts of the Cell

There are three main parts of the cell – plasma membrane, cytoplasm and its organelles and the nucleus

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1. Plasma membrane

- It is also known as the *cell membrane* which is the outermost cell surface, which separates the cell from the external environment.
- It is composed primarily of proteins and lipids, especially phospholipids.

2. Cytoplasm and organelles

- It is also called a **cytosol**, a semiliquid substance that composes the foundation of a cell.
- It is the gel-like material enclosed by the plasma membrane.
- Within the cytoplasm of eukaryote cells are a number of membrane-bound bodies called **organelles** ("little organs") that provide a specialized function within the cell. The following are the different cytoplasmic organelles:

a. Endoplasmic reticulum (ER)

- It is a series of membranes extending throughout the cytoplasm of eukaryotic cells.
- It is the site of protein synthesis in a cell.
- In some places, the ER is studded with submicroscopic bodies called **ribosomes**. This type of ER is referred to as **rough ER**.
- In other places, there are no ribosomes. This type of ER is called **smooth ER**.

b. Golgi body (Golgi apparatus)

- It is a series of flattened sacs, usually curled at the edges.
- In the Golgi body, the cell's proteins and lipids are processed and packaged before being sent to their final destination.

c. Lysosome

- It is composed of powerful digestive enzymes
- It is used to digest waste materials of the cells
- It breaks down particles of food taken into the cell and make the products available for use.
- It is also called the suicide bag of the cell

d. Mitochondrion

- It releases quantities of energy to form ATP (adenosinetriphosphate).
- Because mitochondria are involved in energy release and storage, they are called the "powerhouses of the cells."

e. Ribosomes

- They make protein for the cell.
- Ribosomes exist floating freely in the cytoplasm, and also bound to the endoplasmic reticulum (ER).

f. Chloroplast

- It is the site of photosynthesis.
- Within chloroplasts, energy from the sun is absorbed and transformed into the energy of carbohydrate molecules.
- Plant cells specialized for photosynthesis contain large numbers of chloroplasts, which are green because the chlorophyll pigments within the chloroplasts are green. Leaves of a plant contain numerous chloroplasts.
- Plant cells not specializing in photosynthesis (for example, root cells) have few chloroplasts and are not green.

g. Vacuole

- It is a large fluid-filled central organelle found in mature plant cells
- The vacuole may occupy more than 75 percent of the plant cell.
- In the vacuole, the plant stores nutrients as well as toxic wastes.

h. Cytoskeleton

- It is an interconnected system of fibers, threads, and interwoven molecules that give structure to the cell.
- The main components of the cytoskeleton are microtubules, microfilaments, and intermediate filaments. All are assembled from subunits of protein.

i. Centriole

- It is a cylinder-like structure that occurs in pairs.
- Centrioles function in cell division.

Nucleus

- Based on the nucleus, cells can be classified as to prokaryotic and eukaryotic cells
- Prokaryotic cells lack a true nucleus; the word *prokaryotic* means "primitive nucleus."
- Eukaryotic cells have a distinct nucleus.

Nucleus has the following parts:

- **Chromosomes**, also known as **chromatin** when the linear units are not obvious.
- The **nuclear envelope**, an outer membrane, surrounds the nucleus of a eukaryotic cell. The nuclear envelope is a double membrane, consisting of two lipid layers (similar to the plasma membrane). Pores in the nuclear envelope allow the internal nuclear environment to communicate with the cell cytoplasm.
- Within the nucleus are two or more dense organelles referred to as **nucleoli** (the singular form is *nucleolus*).

Movement through the Plasma Membrane

In order for the cell cytoplasm to communicate with the external environment, materials must be able to move through the plasma membrane. This movement occurs through several mechanisms.

- **Diffusion**
 - It is the movement of molecules from a region of higher concentration to one of lower concentration.
 - This movement occurs because the molecules are constantly colliding with one another.
- **Osmosis**
 - It is the movement of water from a region of higher concentration to one of lower concentration.
 - It often occurs across a membrane that is semipermeable. A semipermeable membrane lets only certain molecules pass through while keeping other molecules out.
 - This process takes place in living cell with cell membrane as the semipermeable membrane.
 - The effect of the direction of osmosis on the living cell can be demonstrated when cell is placed in different types of solutions such as hypotonic solution, hypertonic solution and isotonic solution
 - A hypotonic solution contains lower concentration of dissolved materials or solute than the cell, making the concentration of water in the solution greater than in the cell. The cell swells when placed in this kind of solution because water enters the cell.
 - A hypertonic solution contains higher concentration of solutes than the cell, making concentration of water in the solution lower than the cell. The cell shrinks when placed in a hypertonic solution because the water moves out of the cell. The shrinking of cell is called plasmolysis.

- Isotonic solution has equal concentration of solute and water with the cell. The cell neither swell nor shrink when placed in an isotonic solution because the cell gains and losses equal amounts of water.
- **Facilitated diffusion**
 - Certain proteins in the membrane assist facilitated diffusion by permitting only certain molecules to pass across the membrane.
 - The proteins encourage movement in the direction that diffusion would normally take place, from a region with a higher concentration of molecules to a region of lower concentration.
- **Active transport**
 - During active transport, a protein moves a certain material across the membrane from a region of lower concentration to a region of higher concentration.
 - Since this movement is happening against the concentration gradient, the cell must expend energy that is usually derived from a substance called adenosine triphosphate or ATP. One example of active transport occurs in human nerve cells. In this process, sodium ions are constantly transported out of the cell into the external fluid bathing the cell, a region of high concentration of sodium.
- **Endocytosis**
 - It is a process in which a small patch of plasma membrane encloses particles or tiny volumes of fluid that are at or near the cell surface.
 - The membrane enclosure then sinks into the cytoplasm and pinches off from the membrane, forming a vesicle that moves into the cytoplasm.
 - It is of two kinds: phagocytosis and pinocytosis. When the vesicle contains particulate matter, the process is called **phagocytosis**. When the vesicle contains droplets of fluid, the process is called **pinocytosis**.

Plants

Organisms that possess the following characteristics:

- multicellular,
- photosynthetic
- stationary
- slow growth processes
- complex life cycle of alternating generations of sexual sporophyte and sexual gametophyte

The plant body is divided into two main parts:

1. Roots

- The structure that anchors the plant to the ground and absorbs water and minerals from the surrounding environment

- The descending portion of the plant body, generally located below ground

2. Shoots

- The aboveground plant structure that includes the **stem**, which is a framework for leaves, flowers, and fruits
- The stem acts as the structural support for the plant and provides the framework for leaves, flowers, and fruits. The stem contains vascular tissues – the xylem and the phloem. **Xylem** conveys water and minerals up from the roots; **phloem** conveys the products of photosynthesis down from the leaves.
- Leaves serve the primary function as the principal sites of photosynthetic reactions. Leaves have a waxy covering on their upper surface which helps prevent loss of water to the air. Leaves have tiny pores known as **stomates** on the surface that allow substances such as water, oxygen, and carbon dioxide to pass through as they either enter or leave the plant. The evaporation of water vapor through stomata in the leaves is known as **transpiration**.
- Flowers are exclusively produced by plants in the **angiosperm** phyla. It is where fertilization among angiosperms occurs.

Animals

Organisms that possess the following characteristics:

- They have eukaryotic cells, meaning, there is true nucleus because of the presence of the nuclear membrane. The nuclear membrane separates the nuclear materials from the cytoplasm.
- The cells are encased only in the plasma membrane. Animals cells do not have the cell wall.
- Animals are multicellular. Their cells form tissue that may be organized into organs and organ systems.
- Animals for the most part are capable of movement; this property has allowed animals to be distributed throughout the land as well as marine habitats around the world.
- Animals are heterotrophs which means they obtain their material and energy needs by feeding on other organisms.
- They are aerobic which means that they require oxygen when they harness energy from the food.
- The survival of animal species is accomplished through sexual reproduction, where specialized organs called gonads (female ovary/male testis) through meiosis produce genetically-diverse gametes (female eggs/male sperm). However, there are also animals that can reproduce asexually.

III. Physical Sciences

Physical Sciences - focus on the nature and behavior of matter and energy. The following are some of the branches of Physical Sciences:

- **Physics:** The study of matter and energy and the interactions between them.
- **Chemistry:** Deals with the composition, properties, reactions, and the structure of matter.
- **Astronomy:** The study of the universe beyond the Earth

Matter**What is matter?**

- Matter is anything that occupies space and has mass.
- It is made up of tiny particles that are in constant motion. These particles are called atoms.
- Matter can be described based on its characteristics which can either be general or unique.
- General characteristics are shared by a great number of materials. This property does not make an object unique. The following are the general characteristics of matter:
 - **Mass**
 - The amount of matter contained in an object
 - The mass of an object depends on the amount of matter it contains
 - Mass of an object does not change with the change of location. Example, a kilogram of rice will not change whether it is on top of the mountain or at the sea level.
 - Mass is expressed in kg, g, tons, pounds
 - **Weight**
 - It is the gravitational force on an object
 - The weight of an object is the gravitational force exerted by a planet on it
 - The weight of an object depends on the mass and its distance from the center of the planet
 - The weight of an object is expressed in Newton



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- To determine the weight of an object, mass of an object is multiplied to the acceleration due to gravity which has a value of 10m/s^2 ($W = mg$)

○ **Volume**

- The exact space occupied by an object
- No two objects can occupy the same space at the same time. This property is called impenetrability
- Unique characteristics of matter are used to identify them. It is also called properties.
- The following are the unique characteristics or properties of matter:

○ **Density**

- It is the amount of matter in a given volume of material.
- It is expressed in gram per cubic centimeter (g/cm^3)
- Density can be determined using the formula $D = m/v$
- Every substance has a specific density that can be used to identify such substance
- The difference in density of two objects will cause an object to float or sink. Objects that are denser than water sink while an object that is less in water floats in water
- Balloons float because the helium (gas in balloon) is less dense than the air.

○ **Melting Point**

- The temperature at which a solid turns to liquid.
- To change solid to liquid requires the addition of heat.
- When a solid melts, heat causes the atoms to vibrate that the molecular forces holding the atoms together are disrupted.
- It is a unique property that it is a good way to identify a substance.
- Addition of impurities to a substance lowers the melting point of that substance.

○ **Boiling Point**

- The temperature at which a liquid turns to gas.

- To change liquid to gas requires addition of heat.
- Addition of impurities to a substance increases its boiling point.

Some Unique Properties of Metals:

- Thermal and Electrical conductivity
 - They are good conductors of heat and electricity
- Malleability
 - Metals can be pounded and shaped into very thin sheets without breaking
- Ductility
 - The ability of the metals to be drawn into fine wires

What makes up matter?

- It is made up of tiny particles that are in constant motion. These particles are called atoms.
- Atoms are basic building blocks of matter.

What is an atom?

- The smallest particle that makes up any material.
- The word atom is derived from the Greek word *atomos* which means indivisible.
- Atoms are composed of three types of particles: protons, neutrons, and electrons.
- Both the protons and neutrons reside in the nucleus. Protons have a positive (+) charge, neutrons have no charge—they are neutral. Electrons reside in orbitals around the nucleus. They have a negative charge (-).
- Protons and neutrons are responsible for most of the atomic mass.
- It is the number of protons that determines the atomic number.
- The number of protons in an element is constant but neutron number may vary, so mass number (protons + neutrons) may vary.
- The same element may contain varying numbers of neutrons; these forms of an element are called isotopes. The chemical properties of isotopes are the same, although the physical properties of some isotopes may be different.
- All atoms would like to attain electron configurations like noble gases. That means they have completed outer shells.



- Completed outer shell is having eight electrons in the outermost shell which means they are stable.
- Atoms can form stable electron configurations like noble gases by:
 - losing electrons
 - sharing electrons
 - gaining electrons.
- Atoms that have 1, 2 or 3 electrons in their outer levels will tend to lose them in interactions with atoms that have 5, 6 or 7 electrons in their outer levels.
- Atoms that have 5, 6 or 7 electrons in their outer levels will tend to gain electrons from atoms with 1, 2 or 3 electrons in their outer levels.
- Atoms that have 4 electrons in the outer most energy level will tend neither to totally lose nor totally gain electrons during interactions.
- Electrons are arranged in the different energy level, sublevel and orbitals. The arrangement of electrons in different orbitals is known as electron configuration.
- The electronic configuration is expressed by indicating the principal quantum number and its respective orbital along with the number of electrons present in it.
- In the distribution of electrons, the electrons in an atom are arranged in such a way that they occupy orbitals in the order of their increasing energy. The orbital with the lowest energy will be filled first. The orbitals should be filled up using this pattern
1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s.

How are matter classified?

Matter is classified based on the kind of atoms that make up the materials. Based on the atoms that compose them, matter can either be a **pure substance** or a **mixture**.

Pure substances

- The percentage composition is always the same
- The component particles are identical
- It is always homogenous which means having uniform properties all throughout
- They can be divided into two kinds – **elements** and **compounds**
- Elements are made up of only one kind of an atom. Every element has unique properties which can be used to identify them. Examples are: oxygen, pure gold, sodium, carbon
- Compounds are pure substances that are made up of two or more elements or two or more kinds of atoms. The smallest particle of a compound is called a molecule. Examples are: water, carbon dioxide, sugar, salt
- Chemical formula is a short way of naming and writing a compound.

Mixtures

- It is a combination of two pure substances
- Mixtures can either be homogenous or heterogeneous
- Homogenous mixture appears as one uniform material. It has one distinct phase. It can either be a **solution** or a **colloid**.
- Solution is always consists of two parts, the solute and the solvent. The part that dissolves is a solvent and the one that is dissolved is a solute. The solute particles of a solution are evenly spread throughout the solvent. Examples of solution are: vinegar, rubbing alcohol, salt solution.
- Concentration of a solution is expressed in terms of percentage. If the rubbing alcohol is 30% isopropyl, it means that in 100ml rubbing alcohol, only 30ml is isopropyl and 70ml is water.
- A colloid is a homogenous mixture with component particles bigger than those in the solution. Examples are: blood, milk, smoke, fog
- Heterogeneous mixtures do not appear uniform. The component particles can be identified from each other. Examples are: soil, halo-halo, and salad. Suspension is an example of a heterogeneous mixture. Some medicines are in the form of suspension.

How does matter change?

There are three changes of matter: **physical, chemical, and nuclear**.

Physical Change

- Materials change in size, shape or phase.
- No new substance is formed
- The materials retain their own identity.
- A change in phase is a physical change.
 - Melting
 - a change from solid to liquid
 - requires an application of heat

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- Evaporation
 - a change from liquid to gaseous phase
 - requires an application of heat
- Condensation
 - a change from gas to liquid phase
 - requires removal of heat
- Freezing
 - a change from liquid to solid phase
 - heat is removed
- Sublimation
 - a change from solid to gas without passing through the liquid state
 - requires application of heat

Chemical Change

- The material changes its identity because of the rearrangement of atoms, removal of atoms from the molecule, or addition of atoms in the molecule
- New substance is formed.
- The following are signs that a chemical change has taken place:
 - Evolution of light and heat
 - Release of gas
 - Formation of precipitate
 - Production of electrical energy, sound energy and mechanical energy

Nuclear Change

- The nuclei of atoms undergo changes and the atoms becomes transformed into another element
- The atoms themselves change their identities
- During nuclear change, atoms may give off three types of rays:
 - Alpha particles which are the same as the nuclei of a helium atoms. When an atom emits alpha particles, atomic mass of an atom decrease by 4 and atomic number decrease by 2.
 - Beta particles which when emitted by an atom results to a decrease in the atomic number of an atom by 1.
 - Gamma particles which is a form of light with zero mass and no charge.

Energy

What is energy?

- It comes from the Greek word *energeia* which means "in work"
- It makes change possible
- It is used it to do things for us.
 - It moves cars along the road and boats over the water.
 - It bakes a cake in the oven and keeps ice frozen in the freezer.
 - It plays our favorite songs on the radio and lights our homes.
 - It is needed for our bodies to grow and it allows our minds to think.
- It is defined as the ability to do work.
- Energy is found in different forms but they can all be put into two categories:



potential and kinetic. The different forms of energy can also be transformed from one form to another. It is called energy transformation.

- There are two categories of energy- kinetic energy and potential energy. Kinetic energy is energy possessed by an object in motion while potential energy is energy at rest.

Heat Energy

- Heat energy (or just heat) is a form of energy which transfers among particles in a substance (or system) by means of kinetic energy of those particles. In other words, under kinetic theory, the heat is transferred by particles bouncing into each other.
- Heat transfer is a process by which internal energy from one substance transfers to another substance.
- Under the kinetic theory, the internal energy of a substance is generated from the motion of individual atoms or molecules.
- Heat energy is the form of energy which transfers this energy from one body or system to another.

How is heat transferred?

There are different ways by which heat energy be transferred to another substance.

Conduction

- Conduction is when heat flows through a heated solid.
- It occurs when two objects at different temperatures are in contact with each other.
- Heat flows from the warmer to the cooler object until they are both at the same temperature.
- It is the movement of heat through a substance by the collision of molecules. At the place where the two objects touch, the faster-moving molecules of the warmer object collide with the slower moving molecules of the cooler object. As they collide, the faster molecules give up some of their energy to the slower molecules.
- The slower molecules gain more thermal energy and collide with other molecules in the cooler object. This process continues until heat energy from the warmer object spreads throughout the cooler object.
- Some substances conduct heat more easily than others. Solids are better conductors than liquids and liquids are better conductors than gases.
- Metals are very good conductors of heat, while air is very poor conductor of heat.

Convection

- Convection is when heated particles transfer heat to another substance, such as cooking something in boiling water.
- In liquids and gases, convection is usually the most efficient way to transfer heat.
- Convection occurs when warmer areas of a liquid or gas rise to cooler areas in the liquid or gas.
- As this happens, cooler liquid or gas takes the place of the warmer areas which have risen higher.
- This cycle results in a continuous circulation pattern and heat is transferred to cooler areas.

- Heat energy is transferred by the circulation of the air.

Radiation

- Radiation is when heat is transferred through electromagnetic waves, such as from the sun.
- It can transfer heat through empty space, while the other two methods require some form of matter-on-matter contact.
- It is a method of heat transfer that does not rely upon any contact between the heat source and the heated object. For example, we feel heat from the sun even though we are not touching it.
- Heat can be transmitted through empty space by thermal radiation.
- Radiation is a form of energy transport consisting of electromagnetic waves traveling at the speed of light.
- No mass is exchanged and no medium is required.

Light Energy

- Light is a form of radiant energy that can be detected with the eyes.
- Light energy comes from chemical energy, electrical energy and nuclear energy.
- It is a combination of electrical and magnetic energy that travels very, very fast.
- It is estimated to travel at approximately 300 000 km per second.
- It can pass through anything that is transparent, sort of passes through translucent objects (frosted window) but doesn't make it through opaque objects such as a brick wall.
- Light comes from many sources such as fire, light bulbs, and fire flies and of course the sun.
- Eyes are sensitive to light which lies in a very small region of the electromagnetic spectrum labeled "visible light".
- This "visible light" corresponds to a wavelength range of 400 - 700 nanometers (nm) and a color range of violet through red.
- The human eye is not capable of "seeing" radiation with wavelengths outside the visible spectrum.
- The visible colors from shortest to longest wavelength are: violet, blue, green, yellow, orange, and red.
- Ultraviolet radiation has a shorter wavelength than the visible violet light.
- Infrared radiation has a longer wavelength than visible red light.
- The white light is a mixture of the colors of the visible spectrum.
- Black is a total absence of light.

Electromagnetic Spectrum

- It is a continuum of all electromagnetic waves arranged according to frequency and wavelength.
- This spectrum includes not only visible light (only a small portion of this spectrum) but also ultraviolet and infrared, microwave, radio and gamma waves.
- The sun, earth, and other bodies radiate electromagnetic energy of varying wavelengths.



- All electromagnetic energy passes through space at the speed of light (300 million meters per second) in the form of sinusoidal waves.
- Radiation is defined by its wavelength and frequency. Wavelength is the distance between peaks in the wave, i.e. the distance from crest to crest. The frequency is the time interval between passing peaks. Therefore as the wavelength increases, the frequency decreases (and vice-versa).
- The spectrum of waves is divided into sections based on wavelength. The shortest waves are gamma rays, which have wavelengths of 10^{-6} microns or less. The longest waves are radio waves, which have wavelengths of many kilometers. Visible light is a particular band of electromagnetic radiation that can be seen and sensed by the human eye. This energy consists of the narrow portion of the spectrum, from 0.4 microns (blue) to 0.7 microns (red). The infrared range starts at the end of the red spectrum with wavelengths greater than 0.7 microns.

What are the different electromagnetic waves in the electromagnetic spectrum?

- **Radio waves**
 - They are used to transmit radio and television signals.
 - Radio waves have wavelengths that range from less than a centimeter to tens or even hundreds of meters. FM radio waves are shorter than AM radio waves. For example, an FM radio station at 100 on the radio dial (100 megahertz) would have a wavelength of about three meters. An AM station at 750 on the dial (750 kilohertz) uses a wavelength of about 400 meters.
 - Radio waves can also be used to create images. Radio waves with wavelengths of a few centimeters can be transmitted from a satellite or airplane antenna. The reflected waves can be used to form an image of the ground in complete darkness or through clouds.
- **Microwave**
 - Its wavelengths range from approximately one millimeter (the thickness of a pencil lead) to thirty centimeters (about twelve inches).
 - In a microwave oven, the radio waves generated are tuned to frequencies that can be absorbed by the food. The food absorbs the energy and gets warmer. The dish holding the food doesn't absorb a significant amount of energy and stays much cooler.
 - Microwaves are emitted from the Earth, from objects such as cars and planes, and from the atmosphere. These microwaves can be detected to give information, such as the temperature of the object that emitted the microwaves.
- **Infrared**
 - It is the region of the electromagnetic spectrum that extends from the visible region to about one millimeter (in wavelength).
 - Infrared waves include thermal radiation. For example, burning charcoal may not give off light, but it does emit infrared radiation which is felt as heat.
 - Infrared radiation can be measured using electronic detectors and has applications in medicine and in finding heat leaks from houses. Infrared images obtained by sensors in satellites and airplanes can yield important

information on the health of crops and can help us see forest fires even when they are enveloped in an opaque curtain of smoke.

- **Visible Light**

- The rainbow of colors we know as **visible light** is the portion of the electromagnetic spectrum with wavelengths between 400 and 700 billionths of a meter (400 to 700 nanometers).
- It is the part of the electromagnetic spectrum that we see, and coincides with the wavelength of greatest intensity of sunlight.
- Visible waves have great utility for the remote sensing of vegetation and for the identification of different objects by their visible colors.

- **Ultraviolet radiation**

- It has a range of wavelengths from 400 billionths of a meter to about 10 billionths of a meter.
- Sunlight contains ultraviolet waves which can burn your skin. Most of these are blocked by ozone in the Earth's upper atmosphere.
- A small dose of ultraviolet radiation is beneficial to humans, but larger doses cause skin cancer and cataracts.
- Ultraviolet wavelengths are used extensively in astronomical observatories. Some remote sensing observations of the Earth are also concerned with the measurement of ozone.

- **X-rays**

- They are high energy waves which have great penetrating power and are used extensively in medical applications and in inspecting welds.
- X-ray images of our Sun can yield important clues to solar flares and other changes on our Sun that can affect space weather.
- The wavelength range is from about ten billionths of a meter to about 10 trillionths of a meter.

- **Gamma rays**

- They have wavelengths of less than about ten trillionths of a meter.
- They are more penetrating than X-rays.
- Gamma rays are generated by radioactive atoms and in nuclear explosions, and are used in many medical applications. Images of our universe taken in gamma rays have yielded important information on the life and death of stars, and other violent processes in the universe.

1. Plants lose water through transpiration. Which process is responsible for this?

- A. diffusion
- B. capillarity
- C. surface tension
- D. osmosis

Answer: B. Capillarity is the upward movement of water (from roots to leaves) through a very fine tube called xylem. Transpiration is the evaporation of water from the leaves of the plants through the stomates.

2. Which of the following is produced during photosynthesis?

- A. oxygen and inorganic compound
- B. carbon dioxide and water
- C. oxygen and water
- D. oxygen and organic compounds

Answer: D. During photosynthesis, carbon dioxide and water with the aid of sunlight combine producing sugar in the form of glucose and oxygen is released as by product. Glucose is an organic compound.

3. What will be formed when radium isotope, with 88 protons and 138 neutrons undergoes alpha decay?

- A. radon atom (Rn^{222}) with 86 protons
- B. francium atom (Fr^{222}) with 87 protons
- C. actinium atom (Ac^{222}) with 89 protons
- D. thorium atom (Th^{232}) with 90 protons

Answer: A. When an atom undergoes alpha decay, it emits alpha particle which results to a decrease in the atomic number of 2 and atomic mass by 4. Atomic number is the same as the number of protons, thus the new atom has an atomic number of 86 or 86 protons. Atomic mass is the sum of the number of protons and neutrons thus Radium isotope has an atomic mass of 226. Emission of alpha particle results in the decrease of atomic mass by 4. The new atom has an atomic mass of 222.

4. Strontium-90 has a half life of 28 years. When atmospheric concentration of strontium-90 was measured, it was found it to be $\frac{1}{8}$ the amount that was first measured many years ago. How many years have passed since their first measurement?

- | | |
|-------------|-------------|
| A. 12 years | C. 84 years |
| B. 28 years | D. 56 years |

Answer: C. It would take half-lives for Strontium to decay to $\frac{1}{8}$ of its original amount, $\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$. Since one half-life is 28 years, three half lives would be 84 years using the computation $28 \times 3 = 84$.

5. How are ions formed?

- A. by gaining or losing protons
- B. by gaining or losing electrons
- C. by sharing protons
- D. by sharing electrons

Answer: B. Ions are charged particles. It is either positively charged called cation or negatively charged or anion. Cation results when there are excess protons while anion is formed when there are excess electrons. Electrons are located outside the nucleus while protons are inside the nucleus, thus electrons are the one being transferred from one atom to another and not the protons.

6. Which of the following waves of visible light has the highest frequency?

- A. violet
- B. green
- C. indigo
- D. blue

Answer: A. Violet has the shortest wavelength thus, it has the highest frequency.

7. Where is the earth located during a lunar eclipse?

- A. behind the moon
- B. between the sun and the moon
- C. at a point 180 degrees from North
- D. behind the sun

Answer: B. Lunar eclipse occurs when the moon passes behind the Earth so that the Earth blocks the sun's rays from striking the moon. This can occur only when the Sun, Earth, and Moon are aligned, with the Earth in the middle.

8. Which of the following parts can only be seen in plant cell?

- A. mitochondria
- B. a nucleus
- C. lysosomes
- D. a cell wall

Answer: D. Cell wall is the part of the cell that makes the plant rigid.

9. How should conclusions be stated?

- A. a form that can be evaluated by others
- B. that it is completely correct, with no mistakes
- C. that it works with only 1 set of data
- D. that no one can refute the conclusion

Answer: A. Conclusion is not final yet. This can still undergo further investigation.

10. A block of wood is at rest on top of a table. Which of the following is correct?

- A. The block of wood has no inertia.
- B. There is no force acting on the block.
- C. The inertia of the book is equal to the inertia of the table.
- D. The block of wood is in equilibrium.

Answer: D. When an object is not moving, it means that equal forces are acting on it, thus it is said to be in equilibrium.

11. What do you call that property of a moving object to continue moving in a straight line unless acted on by outside force?

- A. acceleration
- B. velocity
- C. inertia
- D. speed

Answer: C. Inertia makes the body at rest remains at rest, and a body in motion to continue moving in a straight line at constant speed, unless a force is applied to it.

12. A car initially at rest accelerates in a straight line at 6 m/s^2 . What will be its speed after 3 seconds?

- A. 18 m/s
- B. 0 m/s
- C. 9 m/s
- D. 2 m/s

Answer: A. Speed is the product of acceleration and time.

13. What do we determine when we specify both the magnitude and the direction of an object?

- A. mass
- B. acceleration
- C. velocity
- D. weight

Answer: C. *Velocity* is a vector measurement of the rate and direction of motion or, the rate and direction of the change in the position of an object.

14. The moon's gravity is $1/6$ of the earth's gravity. What will be the weight of the bowling ball on earth?

- A. $1/6$ that of its weight on the moon
- B. it varies
- C. 6 times that of its weight on moon
- D. equal that of its weight on moon

Answer: C. Weight on earth is always higher than that of the moon because of its greater pull of gravity.

5. A certain element has an atomic number of 92 and an atomic mass number of 238. Which is true of this element?

- A. It has 92 protons, 146 neutrons, and 92 electrons
- B. It has 92 protons, 238 neutrons, and 146 electrons
- C. It has 146 protons, 92 neutrons, and 92 electrons
- D. It has 92 protons, 146 neutrons, and 238 electrons

Answer: A. Atomic number is the same as the number of protons. For a neutral atom, number of protons and electrons is equal. Atomic number is the sum of the number of protons and neutrons, thus to determine neutrons, number of protons should be subtracted from the atomic number.

16. What is formed when elements are chemically combined?

- A. alloys
- B. nuclei
- C. mixtures
- D. compounds

Answer: D. Compounds are matter formed from the chemical reactions or combinations of two or more atoms or elements.

17. What is the formula for determining density of an object?

- A. mass plus volume
- B. mass divided by volume
- C. volume divided by mass
- D. mass times volume

Answer: B. Density is defined as mass per unit volume of an object.

18. Which of the following is true for buoyant force?

- A. It's the force of gravity acting on a submerged object
- B. It depends on the density of the submerged object
- C. It is the net upward force of the surrounding liquid acting on a submerged object
- D. It is the difference between a submerged object's weight and the weight of an equal mass of water

Answer: C. When an object is placed in a fluid, the fluid exerts an upward force on an object. This force is called buoyant force.

19. A block of wood floats in water. What can be said of its buoyant force?

- A. Its buoyant force is greater than the weight of the wood
- B. Its buoyant force is equal to the weight of the wood
- C. Its buoyant force is less than the weight of the wood
- D. Cannot be determined because the block is not completely submerged

Answer: B. A completely submerged body displaces a volume of liquid equal to its own volume. The buoyant force equals the weight of the fluid displaced.

20. An object with a mass of 2 kg displaces 1000 ml of water. Which of the following is true?

- A. The weight of this object is 20 N.
- B. The weight of this object is 5 N.
- C. The buoyant force on this object is 500 N.
- D. The buoyant force on this object is 30 N.

Answer: A. Weight of an object is determined by multiplying the mass of an object and the acceleration due to gravity which is equal to 10m/s^2 .

21. An object with a mass of 1.5kg displaces 1kg of water. Which of the following is true?

- A. The buoyant force of this object is 10N.
- B. The buoyant force of this object is 15N.
- C. The buoyant force of this object is 25N.
- D. The density of the object is less than that of water.

Answer: A. The buoyant force equals the weight of the fluid displaced.

22. Water has a higher specific heat than iron. What does this mean?

- A. water is hotter than iron
- B. water heats more rapidly than iron
- C. water is more dense than iron
- D. water heats more slowly than iron

Answer: D. Specific heat capacity of a substance is defined as the heat required to raise unit mass of substance by one degree of temperature.

23. Which of the following is responsible for heat transfer by radiation?

- A. atmospheric currents
- B. collision of particles
- C. electromagnetic waves
- D. molecular and electronic collision

Answer: C. Radiation is a type of heat transfer where heat is transferred through electromagnetic waves. An example is the heat transfer using the infrared radiation from the sun.

24. Which of the following is true when an object is positively charged?

- A. It has a deficiency of neutrons
- B. It has an excess of neutrons
- C. It has a deficiency of electrons
- D. It has a deficiency of protons

Answer: C. Electrons are negatively charge particles. When an object is deficient of electrons, it means it has excess protons thus making it positively charge.

25. What do you call the charges in an electrical circuit flowing in the same direction?

- A. constant current
- B. an alternating current
- C. direct current
- D. basic current

Answer: C. DC (direct current) is the unidirectional flow or movement of electric charge carriers which are usually electrons.

26. If the current in a wire is 5 amperes, how much charge will flow through it in 5 second?

- A. 10 coulombs
- B. 25 coulombs
- C. 2.5 coulombs
- D. 2 coulombs

Answer: B. The formula for computing charge is $Q = (I)(t)$, where Q stands for charge expressed in coulombs, t is time in seconds, and I for current expressed amperes.

27. If a voltage of 100 volts produces a current of 5 amperes in an electrical device, what is the resistance?

- A. 95 ohms.
- B. 20 ohms.
- C. 105 ohms.
- D. 500 ohms.

Answer: B. Resistance can be computed using the formula $V = IR$, where V stands for voltage, I for current, and R for resistance. $R = V/I$.

28. If three light bulbs of different wattage are connected in series to a battery, which of the following is true?

- A. the resistance in each light bulb will be the same
- B. the current in each light bulb will be the same
- C. the light output of each bulb will be the same
- D. the power consumed by the light will be the same

Answer: B. The current that flows in a series circuit is the same throughout the entire circuit.

29. What is true when material is transparent?

- A. light can pass freely through it in a straight line
- B. it reflects light
- C. it cannot emit any light
- D. it absorbs light and redistributes the thermal energy

Answer: A. A transparent material one that allows light to pass through with little absorption or distortion. If there is distortion, an object is translucent instead of transparent.

30. What is the weight of an object whose mass is 17.6 g?

- A. 0.176 N
- B. 1.76 N
- C. 176 N
- D. 17.6 N

Answer: A. Weight is determined using the formula $W = mg$, where m is the mass expressed in kilograms and g is the acceleration due to gravity with a value of 10m/s^2 . In the problem above, 17.6 g should be converted first to kg before multiplying to 10. To convert g to kg, divide by 1000.

31. Using the table below, what is the independent variable?

Food use	Temperature	Room Lighting	Exposure	Amount of moisture	Result
White bread	35degrees	Dimly lighted	Air	High	Plenty of molds
White bread	35degrees	Dimly lighted	Air	Low	Tiny spots of mold

- A. type of bread
- B. temperature
- C. exposure time
- D. amount of moist growth

Answer: D. Independent variable is the factor being changed in the experiment. In the above experiment, the amount of moisture is the one changed or altered.

32. What conclusion can be formulated/stated from the above experiment?

- A. Molds grow on bread.
- B. Molds growth is dependent on temperature.
- C. Molds grow best in moist places.
- D. Molds like breads.

Answer: C. The table shows that plenty of molds grow on bread with high moisture.

33. What absorbs a portion of the radiation from the sun, preventing it from reaching the earth's surface?

- A. chlorofluorocarbon
- B. hydroflourocarbon
- C. carbon dioxide
- D. ozone layer

Answer: D. The ozone layer absorbs the ultraviolet radiation from the sun. Ultraviolet is a portion of the sun's rays.

34. Which of the following is true about sub atomic particles, mass number and atomic number?

- A. The number of protons and electron is equal to the mass number.
- B. Atomic number is equal to the number of protons
- C. Mass number is equal to the number of neutron and electron
- D. Mass number and atomic number are equal.

Answer: B. Atomic number is always equal to the number of protons. Atomic mass or mass number is equal to the number of protons and neutrons in the nucleus of an atom.

35. What do you call atoms of the same elements having the same atomic number but have different mass number due to differences in their number of neutrons?

- A. isotopes
- B. inert gases
- C. isomers
- D. ion

Answer: A. Isotopes are defined as atoms of the same element with the same atomic number but different atomic masses.

36. Given the following electron configuration, determine the group number and period number of the element. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$

- A. period 4, group 2
- B. period 2, group 5
- C. period 7, group 20
- D. period 6, group 2

Answer: A. Period of an element is identified/determined by looking at the highest energy level occupied by the electrons. Group is determined by the number of electrons in the last configuration.

37. A chemical bond is an attractive force that holds atoms together. What type of chemical bond is referred to as electrovalent or electrostatic attraction between positive and negative ions?

- A. metallic bond
- B. abnormal bond
- C. ionic bond
- D. covalent bond

Answer: C. Ionic bond is a chemical bond formed by the transfer of electrons. Transfer of electrons results in the formation of positive and negative ions thus ionic bond is also called electrostatic bond because of the attraction between positive on and negative ion.

38. Which of the following statement is true?

- A. orbitals make up a subshell; subshells make up a shell
- B. shells make up an orbital; orbitals make up a subshell
- C. subshells make up a shell; shells make up an orbital
- D. shells make up subshell; shells make up orbital

Answer: A. Shells or the main energy level are made up of subshells or sub energy levels. Subshells are made up of orbitals.

39. If 100 g of reactants are used up in a reaction, what will be the mass of the products?

- A. 100
- B. 45
- C. 25
- D. 50

Answer: A. According to the Law of Conservation of Mass, the mass of substances produced (products) by a chemical reaction is always equal to the mass of the reacting substances (reactants).

40. Which of the following is not a chemical reaction?

- A. ripening of fruit
- B. rusting of metal
- C. freezing of water
- D. burning of wood

Answer: C. Freezing of water does not involve a change in the composition of the substance. There is no new product/s formed.

41. What happens to the volume of a confined gas if its pressure is doubled and its temperature remains constant?

- A. decrease
- B. will remain the same
- C. increase
- D. impossible to determine

Answer: A. According to Boyle's Law, volume of the gas is inversely proportional to the pressure when the temperature is constant. Since the pressure is doubled, it is understood that the volume decreased.

42. What differentiates a eukaryotic cell from a prokaryotic cell?

- A. presence of true nucleus
- B. presence of cell membrane
- C. presence of chromosomes
- D. none of the above

Answer: A. Eukaryotic cells have true nucleus because of the presence of the nuclear membrane which separates the nuclear parts from the cytoplasmic organelles.

43. What best describes a cold blooded animal?

- A. They thrive in cold environment
- B. Their body fluid temperature adapts to environment
- C. They first order consumer
- D. Their blood is cold

Answer: B. Cold blooded animals are organisms that adapt the temperature of the environment.



44. Organisms may reproduce sexually, asexually, or both sexually and asexually. Which of the following is an example of sexual reproduction?

A. vegetative reproduction
B. union of gametes
C. fission
D. budding

Answer: C. Sexual reproduction involves union of sex cells or gametes.

45. During photosynthesis, plants produce food in the form of glucose, known as the end product of photosynthesis. In what form do plants and other autotrophic organisms store food?

A. Root Crops
B. Fruits
C. Rhizome
D. Starch

Answer: D. Plants store food in the form of starch.

46. Genes are known as the units of heredity because they carry the hereditary information from generation to generation. What organic compound in the genes does this work?

A. Carbohydrates
B. Lipids
C. Proteins
D. Nucleic Acid

Answer: D. The DNA in the genes is the one responsible for passing traits from parents to offspring. DNA is a ribonucleic acid.

47. Which of the following is the correct sequence of the levels of organization in human body?

A. cells > tissues > organs > system
B. organs > tissue > system > cells
C. system > organs > tissue > cells
D. None of the above

Answer: A. All living things are made up of cells. Cells are grouped together according to functions and location and this is called tissue. Tissues are grouped together forming organs. Organs are grouped together to form systems.

48. The following are insects except one. Which is not an insect?

A. mosquito
B. firefly
C. spider
D. ant

Answer: C. Spider is an arachnid and not insect. Insects have three pairs of legs while spider has four pairs of legs.

49. Which of the following is true of inertia?

A. It is the tendency of an object at rest to remain at rest.
B. It is the tendency of an object in motion to gain speed.
C. It is the tendency of the moving object to change its direction.
D. It is the tendency of matter to disintegrate.

Answer: A. Inertia is the tendency of an object to remain at rest unless a force is applied to it.

50. Which is the hottest star?

- A. Red
- B. Blue
- C. Yellow
- D. White

Answer: B. Based on the temperature, from the highest to the lowest temperature, stars are arranged this way: blue, white, yellow, and red.

51. Which of the following statements explains Boyle's Law?

- A. The pressure exerted by a certain amount of gas is directly proportional to the volume of that gas as long as the temperature remains constant.
- B. The volume of a certain amount of gas is directly proportional to the temperature of that gas in any situation.
- C. The pressure exerted by a certain amount of gas is inversely proportional to the volume of that gas as long as the temperature remains constant.
- D. The volume of a certain amount of gas is directly proportional to the temperature of that gas as long as the pressure remains constant.

Answer: C. Boyle's Law states that the pressure exerted by a certain amount of gas is inversely proportional to the volume of the gas as long as the temperature remains constant. It means that the pressure that a particular gas exerts will increase as the volume of the gas decreases or the pressure that a particular gas exerts will decrease as the volume of the gas increases, provided that the temperature does not change.

52. Which of the following statements explains Charles' Law?

- A. The pressure exerted by a certain amount of gas is inversely proportional to the volume of that gas as long as the temperature remains constant.
- B. The volume of a certain amount of gas is directly proportional to the temperature of that gas as long as the pressure remains constant.
- C. The volume of a certain amount of gas is directly proportional to the temperature of that gas in any situation.
- D. The volume of a certain amount of gas is inversely proportional to the temperature of that gas as long as the pressure remains constant.

Answer: B. Charles Law states that the volume of a certain amount of gas is directly proportional to the temperature of that gas as long as the pressure remains constant. It means that the volume of a gas will increase as the temperature increases and the volume of gas will decrease as the temperature decreases provided the pressure does not change.

53. Which of the following explains the ideal gas law?

- A. The current state of an ideal gas depends upon the volume, pressure, and temperature of that gas.
- B. The temperature of an ideal gas has no effect on the state of that gas
- C. The volume of a certain gas is inversely proportional to the temperature of that gas as long as the pressure remains constant.
- D. The pressure exerted by a certain amount of gas is directly proportional to the volume of that gas as long as the temperature remains constant.

Answer: A. Ideal gas law states that the current state of an ideal gas depends upon the volume, pressure, and temperature of that gas. Ideal gas law is basically a combination of Boyle's law and Charles' law that simply states that there is a relationship between the volume, pressure, and temperature of a particular ideal gas or a mixture of ideal gases. The relationship between volume, pressure, and temperature can be expressed mathematically using the formula $pV = nRT$, where p refers to the pressure, V refers to the volume, n is the amount of gas present, R is the gas constant, and T is the temperature.

54. Which of the following are the three basic components of an atom?

- A. protons, neutrons, ions
- B. protons, neutrons, electrons
- C. protons, electrons, ions
- D. neutrons, electrons, ions

Answer: B. Atoms are the building blocks of matter. It is very small but it is still made up of much smaller particles which are called subatomic particles. The three main subatomic particles are the protons, electrons and neutrons.

55. $^{14}_6\text{C}$ and $^{12}_6\text{C}$ are examples of carbon

- A. ions
- B. molecule
- C. valences
- D. isotopes

Answer: D. Isotopes are defined as atoms of the same elements with the same atomic number but with different atomic masses.

56. Why does greenhouse effect result to warmer temperature near the surface of the earth?

- A. Ozone traps ultraviolet radiation
- B. Carbon dioxide traps infrared radiation
- C. Clouds traps infrared radiation
- D. Soil absorbs incoming radiation

Answer: B. When sunlight reaches the earth's surface, it is absorbed by rocks, soil, and other objects and is converted to infrared or heat. This energy is radiated by the earth's surface back to the atmosphere. Carbon dioxide which is a greenhouse gas prevents this infrared radiation from escaping to the outer space thus increasing the temperature in the lower atmosphere. The heating of the lower atmosphere is called the greenhouse effect.

57. Which two systems of yours work when you move your leg?

- A. Immune and Excretory
- B. Nervous and Muscular
- C. Digestive and respiratory
- D. Circulatory and Integumentary

Answer: B. The nervous system directs and coordinates all the activities of the body. Muscular system functions for movement.

58. Barometer –Air pressure: _____ - _____

- A. Decibel- Sound
- B. Kilometer- Length
- C. Thermometer- Temperature
- D. Odometer- Smell

Ans. C. Barometer is an instrument use to measure air pressure. In the same way, thermometer is an instrument use to measure temperature.

59. _____ - Cancer: Cardiologist - _____

- A. Geologist – Earth
- B. Cardiologist – Urethra
- C. Oncologists – Heart
- D. Leukemia – Lymph Nodes

Answer: C. Oncologist specializes on cancer while a cardiologist specializes on heart.

60. Which of the following is the function of the chloroplast?

- A. Mitosis and Meiosis
- B. Cellular respiration
- C. Conversion of the light energy to chemical energy
- D. Protein synthesis

Answer: C. Chloroplast is the site of photosynthesis. Photosynthesis is the food making process in plants. During photosynthesis, the plants absorb the light energy from the sun which is converted into chemical energy.

61. It is observed that most leaves have fewer stomata on the upper surface than on the lower surface. A greater number of stomata on the upper surface would result in _____.

- A. closing and opening of stomata on the lower surface
- B. excessive loss of water
- C. faster rate of photosynthesis
- D. reduced transpiration

Answer: B. Stomata are the tiny openings in the leaves where water vapor escapes through the process of transpiration. The upper surface of the leaf of the plants is directly exposed to sunlight thus more stomata on the upper surface would mean higher rate of transpiration or excessive evaporation of water.

62. A student placed some sliced of potato in a glass of drinking water. After a while she took a few slices and found the slices very turgid. What can you infer from this observation?

- A. The slices became smaller
- B. The potato slices did not change in shape
- C. Water was absorbed by the potato
- D. Water moved out of the tissues of potato



Answer: C. The drinking water represents a hypotonic solution. When cells are placed in a hypotonic solution, the water molecules move towards the cell. This is the reason the potato became turgid.

63. Which of the following can contribute to the "greenhouse effect"?

- A. Land plants using carbon dioxide
- B. Depletion of minerals in soil by over farming
- C. Destruction of forest
- D. Capturing radiant energy using solar cells

Answer: C. Greenhouse effect is the warming of the lower layer of the atmosphere. The green house gases such as the carbon dioxide trap the infrared radiation preventing it from radiating back to the space. Plants are the organisms that make use of the carbon dioxide in the process of photosynthesis.

64. In taxonomy, which classification is arranged from largest related groups to the smallest?

- A. Species, genus, order, family, class, phylum, kingdom
- B. Species, order, genus, family, class, phylum, kingdom
- C. Kingdom, Class, Phylum, Order, Family, Genus, Species
- D. Kingdom, Phylum, Class, Order, Family, Genus, Species

Answer: D. The highest level of classification is kingdom. Kingdom is divided into different phyla (phylum). Phylum is divided into different classes. Class is divided into different orders. Order is divided into different families. Family is divided into different genera and Genus into species.

65. Which of these is the correct representation of the hierarchical organization of life from least to most complex?

- A. hydrogen, water, heart muscle cell, nucleus, heart muscle tissue, heart, human
- B. hydrogen, water, nucleus, heart muscle cell, heart muscle tissue, heart, human
- C. hydrogen, water, nucleus, heart muscle cell, heart, heart muscle tissue, human
- D. water, hydrogen, nucleus, heart muscle cell, heart muscle tissue, heart, human

Answer: B. The arrangement is from atom– molecule - nucleus – cell – tissue – organ – organism.

66. Oxygen has an atomic number of 8. Therefore, it must have

- A. 8 protons
- B. 8 electrons
- C. 6 neutrons
- D. Only A and B are correct

Answer: D. Atomic number is the same as the number of protons. A neutral atom contains equal number of protons and electrons.

67. The mass number of an element can be easily approximated by adding together the number of

- A. protons and neutrons
- B. electron orbitals in each energy level
- C. protons and electrons
- D. neutrons and electrons

Answer: A. Mass number is determined by adding the number of protons and neutrons in the nucleus of an atom.

68. A mango leaf is at which level in the hierarchial organization of life?

- A. Tissue
- B. organ
- C. organelle
- D. population

Answer: B. Leaf and all other parts of the plants such the roots, stems, flowers are called the plants organs.

69. Which of the following includes all the others?

- A. ecosystem
- B. community
- C. Individual
- D. population

Answer: A. It is the highest level of organization in the given list.

70. In a natural community, the primary consumers are

- A. herbivores.
- B. carnivores
- C. scavengers
- D. decomposers

Answer: A. Primary consumers are those that feed on the producers. Consumers that feed on producers are also called herbivores.

71. The following activities change the composition of the substance EXCEPT

- A. Corrosion of metal
- B. Boiling of water
- C. Decaying of leaves
- D. Curdling of milk

Answer: B. Boiling of water is a physical change thus no change in the composition of the materials.

72. Which of the following does not show a chemical change?

- A. Formation of precipitates
- B. Evolution of gas
- C. Change in temperature
- D. Change in phase

Answer: D. A change in phase is a physical change. Formation of precipitates, evolution of gases and change in temperature are all signs that a chemical reaction has taken place and therefore substances undergo chemical change.

73. Which of the following is non-biodegradable?

- A. plastic straw
- B. animal bones
- C. tea leaves
- D. dried plants

Answer: A Non- biodegradable materials are products that do not break down naturally. When disposed in a garbage pile, the air, moisture, climate, or soil cannot break them down naturally to be dissolved with the surrounding land. Plastics are examples of non-biodegradable materials.

74. A stone is dropped in a graduated cylinder previously filled with water. This will cause the rising of the level of water in the graduated cylinder. Why does the level of water rise?

- A. The stone has more mass than water
- B. The water is less dense than stone
- C. The stone is heavy
- D. The stone takes the place occupied by water

Answer: D. No two objects can occupy the same space at the same time. This property of matter is called impenetrability. Water rises because the stone takes the place of the water.

75. Which of the following is the manifestation of life?

- A. reproduction
- B. genetic control
- C. responsiveness
- D. all of the above

Answer: D. All the characteristics given are characteristics of life. Living things have the ability to reproduce by asexual means, sexual means or both sexual and asexual. Living things have genetic control. They have the ability to pass on traits from the parents to offspring. Living things are also responsive. They have the ability to react to the changes in the environment.

76. Why is it easier to stay afloat when on a beach than when in a swimming pool?

- A. Freshwater is denser than sea water
- B. Sea water is denser than the freshwater
- C. Freshwater provides a greater buoyant force on you
- D. Beach has a bigger area than a swimming pool

Answer: B. Sea water is denser than fresh water. The weight of the sea water that you displace is greater than the weight of the freshwater that you displace. Sea water provides a greater buoyant force on you thus floating is easier.

77. Which of the following electromagnetic waves has the most penetrating power?

- A. x-rays
- B. radio waves
- C. gamma rays
- D. ultraviolet rays

Answer: C. Gamma rays has the shortest wavelength thus, it has the most penetrating power.

78. Mother needs a material to be used as a handle of a cooking utensil which of the following materials should she use?

- A. Aluminum
- B. Wood
- C. Iron
- D. Copper

Answer: B. An insulator should be used as a handle of a cooking utensil so that heat will not pass through the object. Among the objects in the list, wood is the only insulator. All the other objects are metallic and are good conductors of heat.

79. Which of the following characteristics is true both in pure substances and mixture?

- A. It has a sharp boiling point
- B. It has a definite composition
- C. It consists of a single phase
- D. Its components lose their individual properties.

Answer: C. Having a sharp boiling point is a characteristic of a pure substance but not of a mixture. Mixtures have different boiling points depending on the amount of solute dissolved in a solvent. A definite composition is a property of a pure substance but not of a mixture. A mixture can be prepared in any proportion. In mixture, the components do not lose their individual properties but retain their own properties. Both the pure substance and a mixture appear as single phase.

80. Which of the following results are is not due to depletion of the stratospheric ozone layer?

- A. Higher rates of skin cancer
- B. Higher rates of lung cancer
- C. Greater incidence of premature skin aging
- D. Enhanced incidence of severe sunburns

Answer: B. Ozone traps the ultraviolet radiations from the sun. When ozone layer is depleted, the UV can enter the lower layer of the atmosphere. UV can damage skin and eyes but not lungs.



81. Which of the following best describes the process of osmosis?

- A. It is a is the physical process by which the cytoplasm of a cell divides to split the two cells.
- B. It is a is the physical process by which a cell divides itself into four completely new cells.
- C. It is a is the physical process by which a cell divides itself into two completely new, but identical cells.
- D. It is a is the physical process by which fluid moves through a membrane.

Answer: D. Osmosis is the physical process by which a fluid, such as water, moves through a membrane. During osmosis, a fluid will move through a partially permeable membrane, from an area of higher concentration of solvents to an area with a lower concentration of solvents.

82. Which of the following statements describes the process of metabolism?

- A. It is process by which an organism produces and uses energy.
- B. It is process by which fluid moves through a membrane.
- C. It is process by which a cell duplicates itself.
- D. It is process by which a cell defends the organism from outside attackers.

Answer: A. Metabolism is the process by which an organism produces and uses energy. The process of metabolism consists of a variety of chemical reactions that occur within an organism that each help the organism's body produce and use energy.

83. Which of the following is an alternative to ozone depleting substances?

- | | |
|-----------------------|-------------------------|
| A. Hydroflourocarbons | C. Carbon Tetrachloride |
| B. Chlorofluorocarbon | D. Methyl bromide |

Answer: A. It does not contain bromine and chlorine. Ozone depleting substances are those that contain bromine and chlorine ions.

84. Why is predation an important process in maintaining balance in an ecosystem?

- A. it kills all harmful organisms
- B. it keeps the size of the population
- C. it prevents the small organisms from wandering
- D. it makes the food chain complete

Answer: B. Predator controls the population of its prey.

85. A scientific study showed that the depth at which algae were present in a lake varied in different times. During sunny days, the algae were found as much as 6 meters below the surface of the water. However, on cloudy days but were only 1 meter below the surface. Which hypothesis best explains these observations?

- A. Wind currents affect the growth of algae
- B. Precipitation affects the growth of algae.
- C. Nitrogen concentration affects the growth of algae.
- D. Light intensity affects the growth of algae.

Answer: D. Based on the observation, growth of algae is affected by the intensity of light they receive as indicated by the depth of their location during days of different intensity of light.

86. Which of the following colors have the longest wavelength?

- A. Red
- B. Green
- C. Violet
- D. Orange

Answer: A. The visible light has a wavelength ranging from 400-700 nanometers and a color range from violet to red. From shortest to longest wavelength, the visible colors are arranged in the electromagnetic spectrum in this way: violet, indigo, blue, green, yellow, orange, and red.

87. Acid rain occurs when

- A. carbon dioxide combines with water in the atmosphere
- B. phosphorus-rich water in lakes evaporates to form phosphoric acid
- C. sulfur released in burning fossil fuels combines with water in the atmosphere.
- D. excess hydrogen is released into the atmosphere where ozone is formed.

Answer: C. Acid rain is caused by emissions of sulfur dioxide and nitrogen oxides which react with the water molecules in the atmosphere to produce acids.

88. A certain volcano is made up of alternating layer of lava and fragmental materials. Due to this its eruption style may vary from a relatively quiet eruption to a more explosive type. What is this kind of volcano?

- A. cinder
- B. composite
- C. shield
- D. none of these

Answer: B. Cinder cones are volcanoes constructed of pyroclasts ejected from central vent. Pyroclasts are fragments of rock formed by volcanic explosion. These pyroclasts build up to form steep slopes. Shield volcanoes have gentle slopes because they are built of very fluid basaltic lava. When lava reaches the earth's surface, it spreads out like liquids. Composite volcanoes or stratovolcanoes are made of alternating layers of lava and fragmental material. Their eruption vary from a relatively quiet eruption to a more explosive type.

89. Which of the following differentiates intensity and magnitude?

- A. Intensity measures strength while magnitude measures the energy of an earthquake.
- B. Intensity determines the focus of an earthquake while magnitude determines its epicenter.
- C. There is no difference between an earthquake's intensity and magnitude.
- D. Both a and b are correct.



Answer: A. Intensity refers to the effects and damaged produced by the earthquake. Intensity is expressed using the Rossi-Forel scale and uses a roman numeral to express the scale. The greater the damaged, the higher is the intensity. The magnitude of an earthquake is a measure of the energy released by the earthquake. A seismograph is an instrument used to measure the magnitude of the earthquake. It is expressed using a Richter scale and make use of Hindu Arabic numeral.

90. The following are parts of the circulatory system EXCEPT:

- A. lungs
- B. blood
- C. heart
- D. blood vessel

Answer: A. Lungs are parts of the respiratory system.

91. Which of the following parts of the cell is present only among plants?

- A. cell membrane
- B. nucleus
- C. chloroplast
- D. mitochondrion

Answer: C. Chloroplast contains the coloring pigment called chlorophyll.

92. Cations are generated in the process of

- A. Electron gain
- B. Electron lost
- C. Hybridization
- D. Delocalization

Answer: B. Cations are positively charge ions. They are formed when an atom gives off an electron or when electron is lost by an atom.

93. Which of the following shows work is done when blowing a balloon?

- A. The balloon has potential energy which is chaged to kinetic energy
- B. The balloon is filled with air which has weight and occupies space
- C. The balloon moves outward as the force is exerted on it
- D. The force of gravity on the balloon increases.

Answer: C. Work is defined as the product of the force and the displacement. Work is done when the following are observed: (a) force is being exerted on an object, (b) Object moves. (c) The movement is in the direction of the force.

94. Water boils at a lower temperature on a mountain than on the sea level. Why is this so?

- A. The temperature of the air rises as the height increases
- B. The atmospheric pressure is lower as the height increases
- C. The temperature of the air falls as the height increases
- D. The atmospheric pressure is higher as the height increases.

Answer: B. There is less pressure acting on water.

Natural Science

95. When is the sex of the child determined?

- A. During ovulation period of a woman
- B. When the embryo is implanted in the uterus
- C. After the first few division of the zygote
- D. At the moment of fertilization

Answer: D. The sex of the child is determined at the moment of fertilization. Fertilization is the meeting of sperm cell and an egg cell. A child will be a boy when a sperm cell carrying the Y chromosome fertilizes the egg cell and a female when a sperm cell carrying the X chromosome fertilizes the egg cell.

96. Which of the following statements is not true?

- A. Atoms of isotopes of an element have different numbers of protons .
- B. The nucleus of an atom contains the protons and the neutrons.
- C. The nucleus of an atom is positively charge
- D. Atoms are mostly empty spaces. *so much happy!*

Answer: A. Isotopes of elements have different numbers of neutrons but not protons. Isotopes of elements maintain the same number of protons because they have the same atomic number. *ayaw ko ng to*

97. Which type of volcano is the least explosive?

- A. Shield volcano
- B. Composite
- C. Cinder
- D. Stratovolcano

Answer: A. It emits a very fluid basaltic lava that spreads out the way liquids do. *unlabeled*

98. Which of the following is an arthropod?

- A. snail
- B. mosquito
- C. starfish
- D. jellyfish

Answer: B. Arthropods are organisms with jointed legs. Mosquitoes have jointed legs. *feel at home ma*

99. Which of the following is used in classifying rocks into three main types ?

- A. formation
- B. color
- C. texture
- D. size

Answer: A. The three main types of rocks are classified on how they are formed. Igneous rocks are formed from the solidification of molten rock or magma. Sedimentary rocks are formed from sediments of preexisting rocks. Metamorphic rocks are formed when a rock is subjected to tremendous heat and great pressure. *also*

100. Is CO₂ matter?

- A. No, because it is a gas
- B. Not sure, because it can't be seen
- C. Yes, because molecules are matter
- D. Both A and B

Answer: C. Carbon dioxide is a molecule.

